OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

STATION:	SALEM
SYSTEM:	Emergency Operating Procedures
TASK:	Perform Actions for Containment Sump blockage
TASK NUMBER:	N1150900501
JPM NUMBER:	15-01 NRC Sim b
ALTERNATE PATH:	[X]

	RO	SRO
K/A NUMBER:	2.4.4	
IMPORTANCE FACTOR:	4.5	4.7

APPLICABILITY:
EO [ ] RO [X] STA [ ] SRO [X]

EVALUATION SETTING/METHOD: Simulator / Perform

REFERENCES: 2-EOP-LOCA-1, REV. 30, 2-EOP-LOCA-3, Rev. 30, 2-EOP-APPX-7, Rev. 0, OP-AA-101-111-1003, Rev. 6 (All rev checked 5/9/16)

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 5 minutes

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By:	**G Gauding** **Instructor**	**Date: 5-9-16**
Validated By:	**D Cox / M Lutek** **SME or Instructor**	**Date: 8-22-16**
Approved By:	**Training Department**	**Date:** 10/10/16
Approved By:	**Operations Department**	**Date:** 10/10/16

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:

GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

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SIMULATOR SETUP:

RT-1 MALFS:		**VL0120 21SJ44 fails to position (0-100%) Severity 1.0**
		VL0121 22SJ44 fails to position (0-100%) Severity 2.0
I/O's:		**AA04 B LO 21SJ44 SIS SUMP VALVE OPEN: ON**
		AA05 B LO 21SJ44 SIS SUMP VALVE OPEN: ON
		AA05 C LO 22SJ44 SIS SUMP VALVE CLOSED: OFF
		AA04 C LO 21SJ44 SIS SUMP VALVE CLOSED: OFF
ET-1	**KAA01TPN**	**21 RHR pump STOP PB, tied to delete VL0120**
ET-3	**KAA01TPN**	**21 RHR pump STOP PB, tied to delete VL0121**

*****Check SJ44 OPEN lights are lit.*****

Mark up LOCA-3 up to step 21.

INITIAL CONDITIONS:

- A LBLOCA has occurred.
- The crew has responded by performing TRIP-1 and LOCA-1.
- The crew is performing LOCA-3 with all AC buses energized from off-site power, and is waiting at step 21 for RWST lo-lo level alarm.
- All ECCS pumps are operating except 22 CS pump which was stopped at step 8.

INITIATING CUE:
You are the Reactor Operator. Continue performing LOCA-3.

Successful Completion Criteria:
1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:
1. **Secure all ECCS pumps.**
2. **Reestablish recirculation flow with ONE RHR pump and ONE charging OR SI pump.**

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Simulator Operator: Insert RT-1 to cause cavitation of both RHR pumps.		
			Determines ECCS pumps are exhibiting signs of cavitation and recognizes entry into APPX-7 is warranted from Continuous Action Summary. **Evaluator: IF the intention to stop the cavitating RHR pumps is voiced, THEN FOR *RO APPLICANT ONLY*, IMMEDIATELY CUE: The CRS directs you to implement Appendix-7. (This is to prevent activation of Event Trigger when 21 RHR pump is stopped.** **CUE IF REQUIRED**: The CRS directs you to implement Appendix-7. **Note**: Operator may go to EOP-LOCA-5, but first step of LOCA-5 will direct performance of APPX-7. Use above cue if required in LOCA-5.		
			Evaluator: Provide paper copy of 2-EOP-APPX-7.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	APPX-7	CAUTION • Any pump receiving suction from an RHR pump should be stopped before stopping the RHR pump. • If any Charging Pump, SI Pump, or Containment Spray Pump loses suction or shows indication of cavitation, the pump should be stopped.	**Evaluator**: Stopping RHR pumps here **WITHOUT** first stopping any SI or Charging pumps receiving suction from RHR pump discharge is failure criteria.		
	1	(CAS) **MONITOR** RHR Pump suction conditions: a. **CHECK** NO indications of cavitation. • Pump Amps – stable and normal for discharge pressure and flow. • Flow – stable and normal for discharge pressure. • Discharge Pressure – stable and normal for flow • Oscillations – none indicated	Determines BOTH RHR pumps exhibit signs of cavitation, as well as ALL ECCS pumps being supplied from RHR pumps.		

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	1.a RNO	a. **PERFORM** the following:			
		1. **ENSURE** 21RH29 AND 22RH29 in AUTO.	Places 21RH29 and 22RH29 in AUTO.		
*		2. **STOP** ANY Charging Pump or SI Pump taking suction from affected RHR Pump(s)	Stops 21 and 22 Charging pumps and 21 and 22 SI pumps.		
		3. IF indications of cavitation continue, THEN **CLOSE** 21CS36 AND 22CS36, RHR TO CS VALVE.	Determines pump cavitation is continuing, verifies 21CS36 and 22CS36 are shut.		
		4. IF indications of cavitation continue AND radiological conditions permit, THEN **PERFORM** the following: a. **SEND** an operator to release tags and restore CA to RH18(s). b. **CLOSE** affected RH18(s).	**CUE**: Radiation Protection has determined both RHR pump rooms to be unavailable for access due to high radiation levels.		
*		5. IF indications of cavitation continue, THEN **REMOVE** LOCKOUT(s) AND **CLOSE** affected SJ49(s), RHR TO COLD LEG.	Determines RHR pump cavitation is continuing, and removes lockout from 21SJ49 and 22SJ49 RHR TO COLD LEG and shuts 21SJ49 and 22SJ49.		
*		6. IF indications of cavitation continue, THEN **STOP** affected RHR Pump(s).	Determines indications of cavitation continue, and stops 21 and 22 RHR pump. **Simulator Operator: Ensure ET-1 and ET-3 are TRUE when 21 and 22 RHR pump STOP PBs are depressed. This removes the SJ44 failures.**		

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	2	**CHECK** <u>ALL</u> <u>AVAILABLE</u> CFCUs running in Low Speed.	Determines all CFCUs are running in Low Speed.		
*	3.	**SECURE** Containment Spray flow path: a. **STOP** <u>ALL</u> Containment Spray Pumps. b. **CHECK** 21CS36 <u>AND</u> 22CS36 CLOSED.	Stops 21 CS pump. Determines 21CS36 and 22CS36 are shut.		
	4.	(CAS) **CHECK** RWST level >1.2 feet.	Determines RWST level is >1.2 feet.		
	5.	**ATTEMPT** to establish RHR Pump suction: a. **CHECK** 21SJ44 <u>AND</u> 22SJ44 OPEN. b. **CHECK** <u>AT</u> <u>LEAST</u> <u>ONE</u> RHR Pump running, with suction aligned to Containment Sump.	Determines 21SJ44 <u>AND</u> 22SJ44 are open. Determines NEITHER RHR Pumps is running with suction aligned to Containment Sump.		
	5.b RNO	**b. PERFORM** the following:			
	5.b RNO	**1. REMOVE** LOCKOUT(s) <u>AND</u> **CLOSE** affected SJ49(s) RHR DISCH TO COLD LEG.	Determines 21SJ49 and 22SJ49 previously shut.		
	5.b RNO	2. **ENSURE** 21RH29 <u>AND</u> 22RH29 in AUTO	Determines 21RH29 and 22RH29 are in AUTO.		
*	5.b RNO	3. **START** <u>ONE</u> RHR Pump.	Starts 21 OR 22 RHR pump.		

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	5.b RNO	4. IF indications of cavitation occur, THEN **STOP** the running RHR pump, AND **START** the other RHR pump.	Determines cavitation is NOT occurring.		
	5.b RNO	5. IF NO RHR Pumps can be run without cavitation, THEN **STOP** any running RHR Pumps AND **GO TO** Step 8.	Determines RHR pump can be run without cavitation.		
	5.	c) **CHECK** 21RH29 AND 22RH29 in AUTO. d) **REMOVE** LOCKOUT(s) AND **CLOSE** affected SJ49(s) RHR DISCH TO COLD LEG. **e)** **CHECK** ONLY ONE RHR Pump running with suction from Containment Sump.	Determines 21RH29 and 22RH29 are in AUTO. Determines BOTH SJ49's are shut. Determines only one RHR pumps is running.		
	6.	**ATTEMPT** to establish Charging Pump or SI Pump flow in recirculation mode: a. **CHECK** Charging Pump and SI Pump suction aligned to running RHR pump. b. **CHECK** ONLY ONE Charging Pump OR SI Pump running.	Checks Charging Pump and SI Pump suction aligned to running RHR pump by: a.1. Checks 2SJ67 and 2SJ68 SHUT a.2 Checks 22SJ45 OPEN a.3 Checks 21SJ45 OPEN a.4 Checks 21SJ113 and 22SJ113 OPEN a.5 Checks 2SJ1 and 2SJ2 OPEN a.6 Checks 2SJ30 SHUT Determine NO Charging or SI pumps are running.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	6.b RNO	**START** OR **STOP**, Charging and SI Pumps, as required, to obtain ONLY ONE pump running in recirculation alignment.	Starts ONLY ONE Charging or SI pump. **Note**: Per the NOTE at top of page 3, the intent of Step 6 is to start only a single pump (Charging or SI), and it is preferable to start a Charging pump. SAT performance of Step 6 does not require starting a Charging pump, just one of the 4 available charging or SI pumps.		
	6	c. **CHECK** Charging Pump or SI Pump running in recirculation alignment. d. **GO TO** Step 9	Determines Charging or SI pump is running in recirculation alignment.		
			Terminate JPM when operator goes to Step 9.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

___✓ 1. Task description and number, JPM description and number are identified.

___✓ 2. Knowledge and Abilities (K/A) references are included.

___✓ 3. Performance location specified. (in-plant, control room, or simulator)

___✓ 4. Initial setup conditions are identified.

___✓ 5. Initiating and terminating Cues are properly identified.

___✓ 6. Task standards identified and verified by SME review.

___ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

___✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 2 Date 8.22.16

___ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.

___ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

___ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: D. Co [signature] Date: 8.22.16

SME/Instructor: M. Lutek [signature] Date: 8/22/16

SME/Instructor: ___ Date: ___

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INITIAL CONDITIONS:

- A LBLOCA has occurred.
- The crew has responded by performing TRIP-1 and LOCA-1.
- The crew is performing LOCA-3 with all AC buses energized from off-site power, and is waiting at step 21 for RWST lo-lo level alarm.
- All ECCS pumps are operating except 22 CS pump which was stopped at step 8.

INITIATING CUE:

You are the Reactor Operator. Continue performing LOCA-3.